Aditya Vishal Singh

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Skills

Languages: Python, C/C++, JavaScript, SQL

Data & AI / Tools: PySpark, Pandas, FireDucks, LangChain, LangGraph, PydanticAI, PydanticGraph.

Databases & Platforms: PostgreSQL, MySQL, MongoDB, AWS, Apache Iceberg, Apache Parquet, Amazon S3, Amazon Redshift, Docker, Firebase, Azure Service Bus, EC2, ECS, Lambda, SendGrid, WebSockets

Web & Frameworks: FastAPI, Node.js, Express, React.js, Redux, Cloudinary

Work Experience

Pangea Tech, Bangalore Sept 2024 – Present

Software Developer

- Engineered a large-scale, end-to-end data warehouse on AWS processing data from MongoDB and MySQL data lake. Implemented Medallion architecture delivering analytical datasets which sit in the AWS redshift.
- Designed a reliable state-management and checkpointing system to ensure ETL integrity to recovery from partial failures.
- Implemented a Quality Control Agent to automatically detect source schema changes and report downstream impacts over ETL layers, improving pipeline robustness and reducing the manual check before every ETL run.
- Enabled the business to identify recent programs causing $3M in losses and to evaluate downstream partner-of-partner performance, enabling prioritized remediation.
- Built a scalable notification service using Azure Service Bus (topics/subscriptions) supporting email (SendGrid) and real-time in-app delivery via WebSockets with per-user subscriptions and MongoDB status acknowledgements.
- Created an internal tool to extract the schema from SQL/NoSQL sources with more than 95% accuracy, accelerating pipeline development and enabling downstream text to database query and BI tooling.
- Developed a RAG service that performs chunking, embeddings and uses MongoDB-based retrieval for document answering becoming one of the most used services inside the company.
- Designed secure firebase based authentication supporting user-to-system and system-to-system flows with access/refresh tokens.
- *Stack:* Python, FastAPI, PySpark, Pandas, FireDucks, LangChain, PydanticAI, MongoDB, PostgreSQL, MySQL, AWS, S3, Redshift, EC2, ECS, Lambda, Apache Iceberg, Parquet, Docker, Firebase, Azure.

Education

Motilal Nehru National Institute Of Technology, Allahabad 2020 – 2024
B.Tech in Electrical Engineering ***CPI: 7.81***
Prayagraj, India

New Way Senior Secondary School, Lucknow March 2020
Class XII, CBSE ***95.6%***

New Way Senior Secondary School, Lucknow March 2018
Class X, CBSE ***89.4%***

Project Work

- **MusixON**: Built an on-line music player (Shazam API) with secure Google/email authentication, fast song search, playlist management (create, like) and a custom media player featuring shuffle and repeat.
- **DATEIFY**: Dating app where users can find matches. Uses Cloudinary for image hosting, supports detailed profile customization, and employs profile-matching to create a curated user feed.

Achievements

- Solved **1000+ problems** across LeetCode, Codeforces and other competitive platforms.
- LeetCode — Highest rating: **1753**.
- Codeforces — Highest rating: **1417**.
- Part of the winning team in an intra-school basketball championship.

Position of Responsibilities

- Successfully led the 'Dev or Die' development event at Avishkar 2024, the technical fest of MNNIT Allahabad.
- The mentor in the Student Mentorship Program (SMP) provided academic and professional guidance to junior students.